Introduction

Pedestrian counts in dense neighbourhoods rise and fall twice a day, much like tides. We model hourly foot traffic on 41 shopping corridors as the sum of a commuter wave, a lunchtime swell and a slow evening ebb, and ask how much of the variance that shape explains.

Related Work

Earlier sensor studies in Copenhagen and Melbourne fitted daily profiles per street but treated each corridor independently. Gravity models of retail catchment explain where people go, not when.

Infrared beam counters were mounted at both ends of each corridor and calibrated against 312 hours of manual counts. Days with sensor faults or street closures were dropped, leaving 9,840 corridor-days.

Evaluation

Across 41 corridors the tidal model explained 87 percent of hourly variance, against 64 percent for a per-street daily mean. Residuals were largest on market days and during school holidays.